



INDICES & POWERS

Fun with Numbers and Exponents!



WHAT ARE INDICES & POWERS?

Indices (or powers) tell us how many times to multiply a number by itself. It's like a shortcut for repeated multiplication!

For example, 3^2 means $3 \times 3 = 9$. The small number 2 is called the exponent or index.

Another example: 2^3 means $2 \times 2 \times 2 = 8$.

Think of it like this: when you see a big number with a small superscript number above it, that small number tells you how many times to multiply the big number by itself!

This concept makes math easier and helps us write big numbers in a simple way.

VOCABULARY & NOTATION

Understanding the key terms helps us talk about indices and powers correctly. Let's learn what each part of an expression means!



Base:

The number being multiplied (e.g., 3 in 3^2)



Exponent:

How many times we multiply (e.g., 2 in 3^2)



Example:

$3^2 = 3 \times 3 = 9$ (3 is the base, 2 is the exponent)



Also called:

Index or power - all mean the same thing!

UNDERSTANDING POWERS

Powers tell us how many times to multiply a number by itself. Let's explore some fun examples with numbers 2 and 3 to see how powers work!

$$2^1 = 2 \text{ (just the number itself)}$$

$$2^2 = 2 \times 2 = 4$$

$$3^1 = 3 \text{ (any number to power 1 equals itself)}$$

$$2^3 = 2 \times 2 \times 2 = 8 \text{ and } 3^2 = 3 \times 3 = 9$$

PRODUCT LAW

When multiplying powers with the same base, add the exponents! This is called the Product Law of Indices. It makes multiplication much easier.



$$\text{Rule: } a^m \times a^n = a^{(m+n)}$$

$$\text{Example: } 2^2 \times 2^3 = 2^{(2+3)} = 2^5 = 32$$

$$\text{Try: } 3^3 \times 3^2 = 3^{(3+2)} = 3^5 = 243$$

LAW OF INDICES #2



The Quotient Law helps us when dividing powers with the same base. Instead of doing long division, we simply subtract the exponents! The rule is: $a^m \div a^n = a^{(m-n)}$, where m is greater than n .

Rule: $a^m \div a^n = a^{(m-n)}$

Example: $5^4 \div 5^2 = 5^{(4-2)} = 5^2 = 25$

Remember: The base must be the same, and m must be greater than n !

LAW OF INDICES #3

Power of a Power: When a power is raised to another power, multiply the exponents!



Rule: $(a^m)^n = a^{(m \times n)}$

Example: $(3^2)^3 = 3^{(2 \times 3)} = 3^6 = 729$

Think of it like doing the multiplication m times, n times each!

LAW OF INDICES #4 – ZERO EXPONENT LAW

Any number raised to the power of zero equals 1 (where the number is not zero).

Rule: $a^0 = 1$ (where $a \neq 0$)

This might seem surprising at first! Let's see why:

$$7^0 = 1$$

$$10^0 = 1$$

$$5^0 = 1$$

Think about it: when you multiply by a number repeatedly, each step multiplies by that number. But when you stop multiplying (power of 0), you're left with just 1.

Remember: This rule only works if your base number is not zero! 0^0 is undefined.

LAW OF INDICES #5 – ONE EXPONENT LAW

The One Exponent Law is super simple! It says that any number to the power of 1 equals the number itself.

Rule: $a^1 = a$

Example: $9^1 = 9$

This means you don't need to multiply anything - the number stays the same! It might seem too easy, but this law is always true.

Try these on your own:

- $5^1 = 5$
- $100^1 = 100$
- $7^1 = 7$

Remember: Any number with an exponent of 1 is just the number itself!

DIVIDING POWERS

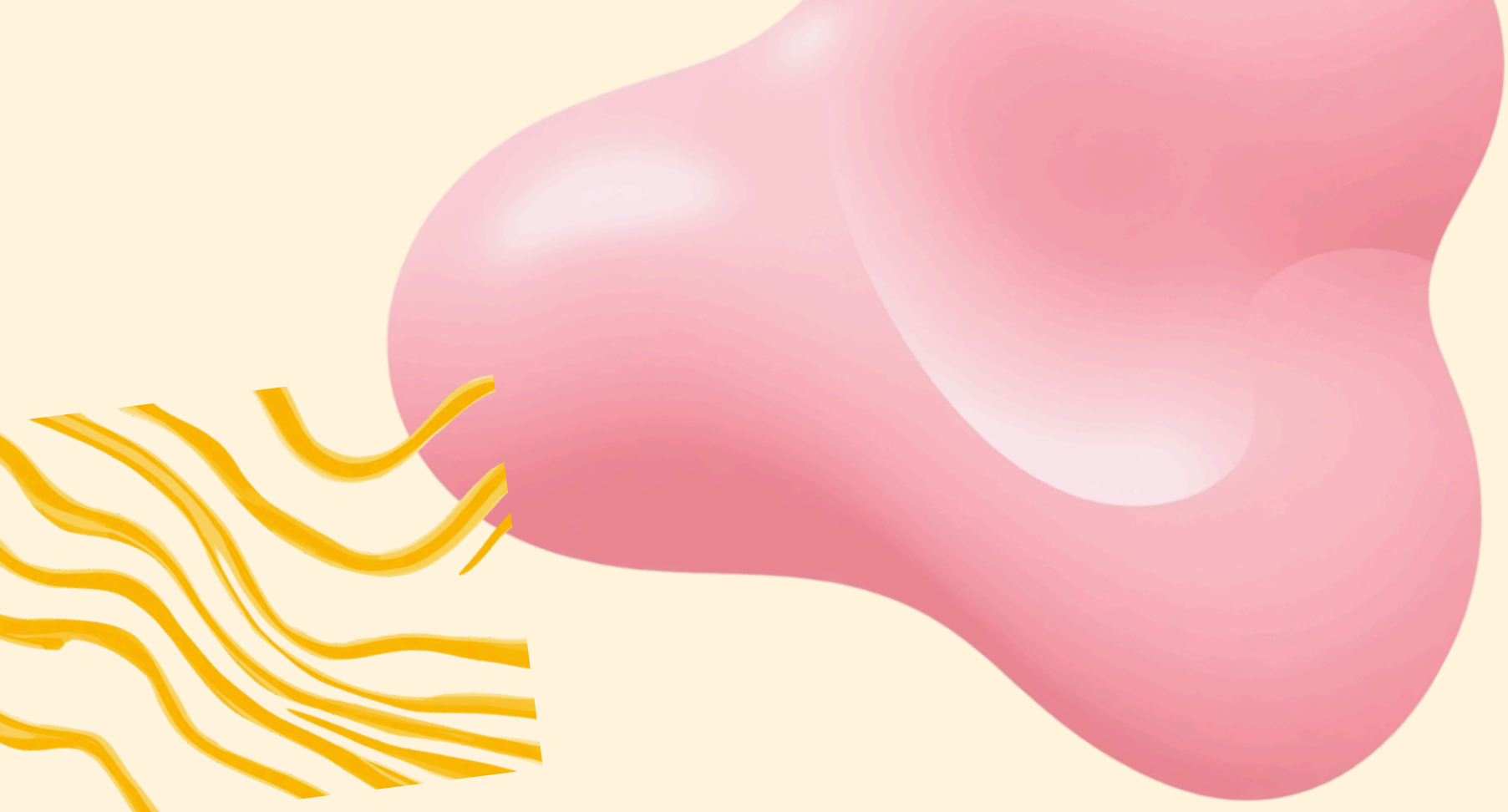


When we divide numbers with the same base, we subtract the exponents! This is called the Quotient Law: $a^m \div a^n = a^{(m-n)}$, as long as m is greater than n .

Example 1: $8^5 \div 8^3 = 8^{(5-3)} = 8^2 = 64$

Example 2: $10^4 \div 10^1 = 10^{(4-1)} = 10^3 = 1,000$

Remember: Always check that the first exponent is bigger!



**THANK YOU
FOR LEARNING
WITH US!**

